

Boundary Dam Integrated CCS Demonstration Project

The World's First Commercial Carbon Capture Unit at a Coal Power Plant

QUICK FACTS

Sector: Power Generation (Coal)

Location: Estevan, Saskatchewan, Canada

Technology: Post-combustion amine scrubbing (SaskPower / Shell Cansolv process)

1. Background and Context

The Boundary Dam power station, operated by SaskPower (Saskatchewan's provincial electricity utility), has provided coal-fired electricity to Saskatchewan since the 1950s. The province's landlocked geography and coal-rich geology made coal the backbone of its electricity system for decades. By the early 2010s, however, Unit 3 of the Boundary Dam station was nearing end-of-life and required either retirement or major refurbishment.

Rather than simply refurbishing the boiler and turbine, SaskPower made an audacious decision: it would integrate commercial-scale post-combustion carbon capture technology into the refurbishment. The project, known as BD3, received CAD 240 million from the Canadian federal government and CAD 110 million from Saskatchewan province, with SaskPower financing the remainder of the CAD 1.5 billion total project cost.

Operator	SaskPower (Saskatchewan Crown corporation)
Location	Estevan, Saskatchewan, Canada (49°N, 103°W)
Commissioned	October 2014 — first of its kind globally
Coal Unit Capacity	139 MW (net after capture energy penalty)
Designed Capture Capacity	1 million tonnes CO ₂ per year
Actual Annual Capture	~0.6–0.8 Mt/year (post-optimisation)
Capture Technology	Shell Cansolv post-combustion amine scrubbing
CO₂ Destination	Weyburn oilfield (EOR) + Aquistore geological storage + food-grade CO ₂

Total CO ₂ Captured (to 2024)	~5 million tonnes
Total Project Cost	CAD 1.5 billion (~USD 1.1 billion)

2. Technology: How Post-Combustion Capture Works at Boundary Dam

The BD3 capture system uses the Shell Cansolv DC-103 amine solvent process. The key steps are:

- **Flue gas pre-treatment:** Before entering the capture unit, the coal plant's flue gas passes through a selective catalytic reducer (for NO_x removal), a fabric filter (particulates), and a flue gas desulphurisation (FGD) scrubber. Sulphur dioxide must be removed before amine scrubbing because SO₂ reacts with amines and degrades the solvent.
- **CO₂ absorption:** Cleaned flue gas enters the bottom of an absorber column and flows upward through a downward-flowing stream of DC-103 amine solvent. The amine selectively absorbs CO₂, producing a 'rich' solvent loaded with CO₂.
- **Solvent regeneration:** The rich solvent is heated in a steam-fed regenerator, releasing a concentrated CO₂ stream (>99% purity). The regenerated 'lean' solvent is recycled to the absorber.
- **CO₂ compression:** The CO₂ stream is compressed to 15 MPa (supercritical conditions) for pipeline transport to the Weyburn oilfield, 66 km away.

Key technical insight: The energy required to regenerate the amine solvent — steam drawn from the power plant's steam cycle — is the main 'energy penalty' of post-combustion CCS. At Boundary Dam, this reduces net power output by approximately 20-25 MW, equivalent to removing roughly 15% of generating capacity from sale.

3. Performance: Design vs. Reality

Metric	Design Target	Actual Performance (Best Year)	Gap and Reason
Annual CO ₂ capture	1.0 Mt/year	~0.80 Mt/year	~20% gap; planned maintenance + system tuning
Capture rate	90%	~87–91%	Within range; varies with load
Solvent degradation rate	Low	Higher than expected	Flue gas contaminants fouled solvent faster
Energy penalty	25 MW	~22–28 MW	Within range; varies with ambient temperature
Plant availability	~95%	~85–90%	Capture unit maintenance causes outages

Metric	Design Target	Actual Performance (Best Year)	Gap and Reason
Cost per tonne CO ₂ captured	~\$80	~\$110–130	Higher operating costs than projected

4. Challenges and What They Revealed

Challenge 1: Solvent Fouling

Contaminants in the flue gas — particularly oxidised sulphur compounds and heavy metal traces — degraded the amine solvent at higher-than-expected rates. This required more frequent solvent replenishment (a significant operating cost) and periodic cleaning shutdowns that reduced annual capture volumes. This finding drove significant R&D; investment into more fouling-resistant solvent formulations — work that benefits all future post-combustion CCS projects.

Challenge 2: Steam Demand Variability

The interaction between the capture unit's steam demand (for solvent regeneration) and the power plant's turbine steam cycle was more complex than models predicted. At certain operating conditions, the capture system's steam demand caused unexpected drops in the power plant's electrical output. This required operational optimisation that took two years to resolve.

Challenge 3: CO₂ Market Dependence

Boundary Dam's business model depended on selling captured CO₂ to the Weyburn EOR operation (which uses CO₂ to extract residual oil) and to a food-grade CO₂ supplier. When demand from either buyer fluctuated, the capture unit had to throttle back, reducing annual totals. This exposed the danger of CCS projects dependent on a single or small number of CO₂ buyers.

5. Achievements and Legacy

Despite performance shortfalls against design targets, Boundary Dam's legacy is profoundly positive for the global CCS community. It proved that commercial-scale post-combustion capture is technically feasible on an operating coal power plant. Its detailed performance data — shared openly through the Global CCS Institute and IEA Greenhouse Gas R&D; Programme — has informed the design of every subsequent commercial CCS project. The solvent fouling data alone has catalysed a generation of improved solvent chemistries.

Achievement	Detail
World-first	First commercial post-combustion CCS at a power plant — October 2014
CO ₂ captured	~5 million tonnes stored or utilised by 2024
Technology validation	Shell Cansolv process validated at commercial scale
Data sharing	Detailed performance data published; informed global CCS design
Cost learning	Demonstrated where cost reductions are possible in future projects

Achievement	Detail
Aquistore	Adjacent deep saline aquifer storage pilot — pure geological storage R&D;

6. Relevance and Lessons for India

- **Scale of India's coal fleet:** India operates over 200 GW of coal-fired power capacity. Many units are relatively recent (built post-2000) and will operate for decades. Boundary Dam shows that CCS retrofit is technically feasible — a critical data point for India's long-term energy strategy.
- **Cost trajectory:** Boundary Dam's per-tonne capture cost (~\$110–130/tonne) was a first-of-a-kind figure. Engineering firms estimate that a similar retrofit project today, benefiting from BD3's operational learnings, would cost \$70–90/tonne — and falling.
- **Revenue model:** India has no equivalent to the Weyburn oilfield for CO₂-EOR. A carbon price mechanism or contract-for-difference is essential to make CCS on Indian power plants economic. The Indian Carbon Market (ICM) and international carbon finance mechanisms (Article 6) are the most likely pathways.
- **Hard-to-abate industries:** India's cement and steel sectors offer stronger economics for CCS than power plants (higher CO₂ concentration in flue gas = lower capture cost). Boundary Dam's experience with power plants sets an upper bound on what industrial CCS will cost.

Key Takeaway for India: Boundary Dam demonstrated the technology. The question for India is not 'can CCS be done?' but 'under what carbon price and policy conditions does it make sense in the Indian industrial context?'